

Memory

Quite simply put memory is of two types, RAM(Random Access Memory) and ROM (Read Only Memory). Both of these are essential in the running of a computer, be it a mobile device or a full fledged desktop.

ROM is the permanent memory on phones it not only consists on memory chips/modules on the device as well as the extensible memory which is fulfilled by means of a memory card. Whereas RAM is purely onboard and is a volatile memory, volatile means that the memory does not retain data upon losing power. They are quite different from each other when it comes to the application of memory. ROM is where data like the applications, images, videos and pretty much everything is stored. The RAM is where data is stored temporarily which needs to be accessed more frequently. So if an application accesses a certain data file way too many times then it might be cached in the RAM to allow faster access, however, this isn't the primary purpose for which RAM exists. Memory is organised in a hierarchy of memory where the fastest is at the top and the slowest at the bottom. The real fast memory are called registers and exist on the processors themselves and hold elements of mathematical equations and such low level data which is needed immediately. Then comes shared memory which allows multiple processors to share memory, after that comes RAM and then ROM.

However, RAM is not cheap compared to other modes of storage and it is volatile which is why not all forms of memory are made using RAM chips. However, Solid State Drives in computers are coming up with chips very similar to RAM but with less volatility which allows for really fast data transfer.

How much RAM do you need?

Most applications these days are fine with mobile devices that have 1 Giga-byte of RAM. However, if past trends are to be studied, more and more applications will soon follow which will need much more RAM. It really comes down to the programmer, if the application uses a lot of elements which constantly change then that much more RAM is needed, otherwise the programmer can make do with ROM. The downside of this approach is that the application becomes slow to the point that casual users can feel the sluggishness of whatever application is under use currently. This does not imply that any sluggish interface is purely due to lack of RAM, there are a lot more factors that come into play here.